

Gowri Das Lovely

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PROFESSIONAL SUMMARY

Systems Engineering graduate student with hands-on experience in model-based systems engineering, requirements engineering, simulation, and machine learning through academic and project-based work. Seeking Systems Engineering and Space Systems internship opportunities.

EDUCATION

Stevens Institute of Technology

Master of Engineering in Systems Engineering; GPA: 3.77/4.00

Hoboken, NJ

Jan. 2026 – Expected May 2027

University of South Wales

Bachelor of Science in Information Communication Technology; Upper Second-Class Honours

Wales, UK

Sept. 2022 – May 2024

Australian Technical and Management College

Diploma in Information Technology

Melbourne, AUS

Sept. 2021 – May 2022

PROJECTS

Autonomous Mars Surface Logistics Rover | Systems Engineering, MBSE, SysML, Gaphor

Spring 2026

- Designed an autonomous Mars logistics rover concept to transport cargo between a surface habitat and lander under communication delays, harsh terrain, dust exposure, and limited maintenance conditions.
- Performed stakeholder analysis, developed operational concepts (CONOPS), system requirements, functional decomposition, interface definitions, and mission phase architecture.
- Created SysML models including Block Definition Diagrams, Internal Block Diagrams, State Machine Diagrams, and Sequence Diagrams using Gaphor.
- Applied House of Quality, Pugh Matrix, N² Diagram, and developed a Verification & Validation strategy covering nominal, degraded, and fault-response scenarios.

Cosmic Object Classification System | Python, PyTorch, CNN, Machine Learning

2025 – Present

- Developed an image-classification pipeline to identify astronomical objects including black holes, red giants, white dwarfs, and nebulae using deep learning techniques.
- Built preprocessing, training, validation, and performance evaluation pipelines using Python, PyTorch, torchvision, pandas, NumPy, PIL, and Matplotlib.
- Compared Convolutional Neural Networks (CNN) and Artificial Neural Networks (ANN) using classification accuracy, confusion matrices, precision, recall, and F1-score.
- Currently expanding the project into a Systems Engineering case study by incorporating stakeholder analysis, system requirements, architecture, interface definition, risk assessment, and verification planning.

Movie Ticket Queue Simulation | Python, SimPy, Discrete-Event Simulation

Spring 2026

- Developed a discrete-event simulation model to analyze customer flow through a multi-stage movie ticketing system using SimPy.
- Modeled customer arrivals, ticket purchasing, concession queues, and service resources to evaluate waiting time, queue lengths, and system throughput.
- Performed simulation experiments under multiple staffing scenarios to identify bottlenecks and recommend operational improvements using statistical performance metrics.

Passport Machine Readable Zone Validation System | Python, Pytest, GitHub Actions

Spring 2026

- Designed and tested software for parsing and validating passport Machine Readable Zone (MRZ) data, including format validation, checksum verification, and error handling.
- Developed comprehensive unit, integration, boundary, mutation, and performance tests using Pytest, unittest.mock, Coverage.py, and automated CI workflows.
- Achieved over 80% code coverage and evaluated system performance using datasets containing more than 10,000 passport records.

EXPERIENCE

Peer Mentor (Selected) | Stevens Institute of Technology

2026 – Present

Hoboken, NJ

- Selected to serve as a Peer Mentor beginning Fall 2026, supporting student transition, campus engagement, and academic success.
- Will provide academic guidance, campus resource navigation, and peer support while fostering student engagement and success.
- Connected students with academic resources, campus services, and support programs to promote academic success.

IT and Administrative Volunteer | The Mentor Ring

2023 – 2024

United Kingdom

- Maintained digital records using Microsoft Excel while supporting documentation, reporting, scheduling, and administrative workflows.
- Provided basic IT support by troubleshooting common computer issues and assisting staff with digital system applications
- Delivered computer literacy guidance to staff and users, improving confidence with everyday digital tools and systems.

Student Ambassador | University of South Wales

2022 – 2023

Wales, United Kingdom

- Represented the university during campus events, student outreach activities, and admissions programs by engaging with students.
- Conducted campus tours, answered student inquiries, and promoted university programs through effective communication.
- Built relationships with prospective students and families by sharing academic experiences and providing guidance on university life.

TECHNICAL SKILLS

Systems Engineering: Requirements Engineering, CONOPS, Functional Decomposition, System Architecture, Verification & Validation, Stakeholder Analysis

MBSE & Simulation: SysML, Gaphor, BDD, IBD, State & Sequence Diagrams

Programming & Data Analysis: Python, SQL, PyTorch, pandas, NumPy, Matplotlib, Flask

Testing & Tools: Pytest, unittest, Git, JMeter, Microsoft Excel